Module 08: Planning — How Agents Think Ahead

> **Course**: Active Inference 101 | **Unit**: Cognitive Science | **Audience**: First-semester undergraduates

Learning Objectives

1. Explain **planning as inference**: how imagining the future is the same process as perceiving the present.
2. Describe **deep temporal models**: how agents think about the future at different timescales.
3. Analyze habits, imagination, and counterfactual reasoning through the Active Inference lens.

Introduction

Animals don't just react to the present — they anticipate the future. A squirrel buries nuts for winter. A chess player thinks three moves ahead. You plan your course schedule for next semester. This module explores how Active Inference explains these remarkable abilities.

The key insight: planning is inference about the future. The same mechanism the brain uses to infer what's happening *now* is used to simulate what *could* happen next.

Key Concepts

1. Planning as Inference

In Active Inference, planning is not a separate cognitive system — it's inference extended into the future:

- **Perception**: "Given my sensory data, what's happening now?" → Minimizing current free energy
- **Planning**: "Given possible action sequences, what *would* happen?" → Minimizing *expected future* free energy

The agent evaluates multiple possible policies (sequences of actions) by imagining their consequences and selecting the policy with the lowest Expected Free Energy (from Module 05).

2. Deep Temporal Models — Thinking on Multiple Timescales

Humans can think about the future at many levels:

- **Milliseconds**: "Where will the ball be when my hand gets there?"
- **Minutes**: "If I leave now, I'll catch the bus."
- **Months**: "If I study hard this semester, I'll get into graduate school."
- **Years**: "If I pursue this career, my life will look like..."

Active Inference handles this through **hierarchical temporal models** — nested layers of prediction, where higher levels make predictions about longer timescales. The top of the hierarchy holds your long-term goals and identity; the bottom handles moment-to-moment movements.

3. Imagination and Counterfactual Thinking

Imagination is "offline" planning — running the generative model in simulation mode:

- "What if I had turned left instead of right?"
- "What would happen if I asked her out?"
- "How would this taste with more salt?"

In Active Inference, imagination uses the same neural machinery as perception, but with sensory input suppressed. This is why imagination can feel vivid, and why dreams (which run the generative model offline during sleep) feel so real.

4. Habits — When Planning Becomes Automatic

For frequently repeated actions, deliberate planning becomes unnecessary:

- The first time you drive to school, you consciously plan each turn
- After months, the route becomes a **habit** — automatic action selection without deliberate evaluation

In Active Inference, habits are policies with such strong priors that they're selected without needing to evaluate EFE each time. They're efficient but inflexible — which is why breaking habits is hard (you must override a strong prior).

5. The Horizon Problem and Bounded Rationality

Planning faces a fundamental challenge: the further ahead you plan, the more possibilities there are. Considering every possible future is computationally impossible. Agents cope by:

- **Pruning**: Only considering a subset of plausible policies
- **Hierarchical decomposition**: Breaking big plans into subgoals
- **Habits**: Automating routine decisions to free up planning capacity for novel situations
- **Satisficing**: Choosing a "good enough" option rather than the optimal one

This connects Active Inference to the psychological concept of **bounded rationality** — the idea that agents are rational given their computational constraints.

Summary

Planning in Active Inference is inference about the future — evaluating possible action sequences by imagining their consequences. Deep temporal models allow agents to plan across timescales from milliseconds to years. Imagination uses the same generative model as perception but in offline simulation mode. Habits provide efficient shortcuts for familiar situations, while deliberate planning handles novel ones.

Further Reading

- Pezzulo, G. (2008). Coordinating with the future: The anticipatory nature of representation. *Minds and Machines*, 18(2), 179-225.
- Friston, K. J. et al. (2021). Sophisticated inference. *Neural Computation*, 33(3), 713-763.
- Dolan, R. J. & Dayan, P. (2013). Goals and habits in the brain. *Neuron*, 80(2), 312-325.